

POOJA SINGH

AUTOSAR MCAL Engineer | Embedded Software Engineer | ADAS | ISO 26262 Functional Safety | ASPICE | Adaptive AUTOSAR | 8+ Years Experience

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PROFESSIONAL SUMMARY

Results-driven Embedded Software Engineer with 8+ years of hands-on experience delivering safety-critical automotive software across AUTOSAR Classic and Adaptive platforms, MCAL development, BSW integration, ADAS, and ECU bring-up. Proven expertise designing low-level drivers in Embedded C and C++ on Infineon AURIX and Renesas RH850 microcontrollers, integrating CAN, CAN-FD, and Automotive Ethernet communication stacks, and executing ISO 26262 (ASIL A-D) and ASPICE Level 2/3 compliant V-model projects. Accelerated ECU integration by 40%, achieved 100% MC/DC coverage, and delivered production software for Volkswagen, AUDI, Renault, and John Deere. Adept at cross-cultural collaboration, mentoring engineers, and driving Agile/Scrum delivery.

CORE COMPETENCIES

AUTOSAR Classic, AUTOSAR Adaptive, MCAL Configuration and Tuning, BSW Stack Integration, ECU Bring-Up and Flashing, ISO 26262 (ASIL A-D), ASPICE SWE.1 to SWE.6, ADAS, Front Camera ECU, CAN, CAN-FD, LIN, Automotive Ethernet, UDS, XCP, DoIP Diagnostics, MISRA-C:2012, Polyspace, LDRA, Agile, Scrum, V-Model SDLC, Cross-Functional Leadership.

TECHNICAL SKILLS

Languages:	Embedded C, C++ (11/14/17), Python, CAPL, Shell Scripting
AUTOSAR:	MCAL, BSW, RTE, SWC, ComStack, NvM, DCM, DEM, FIM, EcuM, OS, Watchdog Manager, ARXML
Standards:	ASPICE SWE.1-SWE.6, ISO 26262 (ASIL A-D), MISRA-C:2012, AUTOSAR 4.x, Polyspace, LDRA
Drivers:	ADC, PWM, DIO, PORT, CAN, SPI, UART, I2C, GPT, WDG, MCU, FlexRay, CDD
MCUs:	Infineon AURIX TC23x, TC27x, TC3xx; ARM Cortex-M/R; Renesas RH850; NXP S32K
Protocols:	CAN, CAN-FD, Automotive Ethernet (DoIP, SOME/IP), LIN, UDS (ISO 14229), XCP, SPI, I2C
Config Tools:	EB Tresos Studio, Vector DaVinci Developer, DaVinci Configurator Pro, Arctic Studio
Debug/Test:	Lauterbach TRACE32, PLS UDE, Vector CANoe, CANalyzer, ETAS INCA, Vector CANdb++
Tools/ALM:	Git, Bitbucket, Jenkins, JIRA, Confluence, DOORS, Polarion, Make, CMake
Methodologies:	Agile (Scrum, Kanban), V-Model, Test-Driven Development, Continuous Integration

PROFESSIONAL EXPERIENCE

Software Engineer - ADAS Front Camera ECU, Magna Electronics GmbH Jun 2024 - Feb 2026
Sailauf, Germany

- Configured and integrated AUTOSAR MCAL and BSW modules (ADC, CAN, SPI, WDG, MCU, PORT, GPT) on Infineon AURIX TC3xx for a safety-critical ADAS Front Camera ECU; executed ECU bring-up, clock tree configuration, and interrupt setup, reducing integration defects by 40%.
- Engineered CAN and CAN-FD communication stack integration and validated end-to-end ECU functionality for Object Detection, Pedestrian Recognition, and Lane Departure Warning features deployed on Volkswagen and AUDI production platforms.
- Resolved 80% of customer-reported defects through root-cause analysis using Lauterbach TRACE32 and PLS UDE, accelerating release cycles and improving sprint velocity by 25%.
- Executed ASPICE SWE.2 to SWE.5 activities; authored Software Detailed Design (SDD), Software Integration Test (SIT), and Software Verification Reports (SVR); achieved 100% MC/DC coverage per ISO 26262 ASIL-B and ASIL-C requirements.
- Collaborated with system, hardware, and validation teams across Germany and India to maintain requirements traceability in DOORS; mentored two junior engineers on AUTOSAR workflows, reducing onboarding time by 50%.

Senior Embedded Software Engineer, Engineering People GmbH Apr 2023 - May 2024
Germany

- Led AUTOSAR BSW configuration and MCAL integration for multiple Tier-1 ECU programs in compliance with ASPICE Level 2/3, MISRA-C:2012 coding standards, and customer architecture guidelines.
- Automated CAPL regression test scripts in Vector CANoe and CANalyzer, reducing manual test effort by 30% and improving defect detection rate by 22%.
- Conducted peer code reviews and walkthroughs across distributed teams, enforcing requirements traceability and Polyspace static analysis to eliminate runtime defects pre-integration.
- Partnered with OEM stakeholders to refine software requirements (SWE.1) and validate detailed design (SWE.2), maintaining alignment with V-model deliverables and customer milestone gates.

Senior Development Engineer - Low-Level Software, Denso Automotive Europe Dec 2022 - Mar 2023
Wegberg, Germany

- Developed Complex Device Drivers (CDD) in Embedded C, owning the full SDLC from SWE.3 (Detailed Design) through SWE.5 (Integration Testing) on safety-critical ECU platforms.
- Diagnosed and resolved OS scheduling and memory-allocation issues using TRACE32, improving runtime stability and reducing CPU load by 18% on Infineon AURIX targets.
- Authored detailed design specifications and integration test cases, ensuring full traceability to system-level requirements per ISO 26262 ASIL-B compliance.

- Spearheaded software development on internal automotive projects compliant with ASPICE Level 2/3, overseeing detailed design artifacts, architecture documentation, and code quality gates.
- Implemented configuration and unit-test frameworks aligned with AUTOSAR 4.x, ensuring deliverables met customer-defined acceptance criteria.

- Integrated AUTOSAR MCAL and BSW modules (COM, CAN, OS, EcuM, DCM, NvM, DEM) for Renault Euro6 ECU platforms using EB Tresos Studio and Vector DaVinci Configurator.
- Generated and reviewed ARXML configuration files, executed change-impact analyses, and supported Hardware-in-Loop (HIL) testing, contributing to a 15% improvement in defect-leakage metrics.
- Delivered ECU software qualification artifacts in an ASPICE-compliant environment, ensuring on-time milestone closure across three concurrent sprint cycles and aligning cross-functional teams across Germany, France, and India.
- Authored unit and integration test cases against OEM coding standards, contributing to compliance audits and supporting customer-side validation campaigns.

- Designed and implemented Embedded C low-level drivers (ADC, PWM, CAN, SPI, UART, DIO) and integrated AUTOSAR MCAL modules on Infineon AURIX TC27x and TC23x for John Deere off-highway ECU platforms.
- Optimized ECU startup sequence across four ECU variants, improving boot performance by 25% and meeting customer-defined wake-up latency targets.
- Led Agile and Scrum sprint planning and retrospective ceremonies for a 6-member engineering team; delivered 95% sprint commitment over 18 sprints and cut build-break incidents by 35% via Git and Jenkins CI workflows.
- Honored with the ERS Champion Award for technical excellence, sprint delivery, and cross-team leadership.

- Architected a Python-based automation tool for one-click DBC and ARXML file generation, eliminating manual configuration steps and reducing turnaround time by 60%.
- Validated CAN communication matrices against OEM specifications, improving delivery consistency and reducing configuration errors by 45% across multiple programs.
- Documented tool usage guides and trained five team members on the automation workflow, ensuring rapid adoption and sustainable knowledge transfer.

- Processed LiDAR and camera sensor datasets for autonomous driving perception systems supporting object detection, scene segmentation, and free-space estimation models.
- Executed unit and integration testing achieving 100% MC/DC code coverage in accordance with ASPICE SWE.4 and SWE.5 requirements.
- Contributed Python and Embedded C verification scripts, strengthening the validation pipeline and improving data-pipeline throughput for ADAS perception stacks.

EDUCATION

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- AUTOSAR Classic Platform - Foundational and Advanced Training (Vector / EB Tresos)
- ISO 26262 Functional Safety - Automotive (ASIL Concepts and V-Model)
- Automotive SPICE (ASPICE) Level 2/3 - Process Awareness
- Embedded C Programming and MISRA-C:2012 Compliance
- Vector CANoe / CANalyzer and CAPL Scripting for Automotive Communication

KEY ACHIEVEMENTS AND AWARDS

- ERS Champion Award, HCL Technologies: Recognized for outstanding technical contribution, sprint delivery, and cross-team leadership.
- Achieved 100% MC/DC code coverage supporting ASPICE SWE.4 and SWE.5 and ISO 26262 ASIL-B and ASIL-C compliance across multiple ADAS programs.
- Delivered production-grade embedded software for Volkswagen, AUDI, Renault, and John Deere across ADAS, powertrain, and off-highway domains.
- Reduced integration defects by 40% and customer-reported issues by 80% through systematic debugging, root-cause analysis, and process improvements.

LANGUAGES

English: Professional Proficiency **Hindi:** Native **German:** Basic (A2)